

Cover letter

Dr. Yanhui Geng is the Director of AI Lab, Huawei Hong Kong Research Centre. Prior to this position, he held the role of Director of Huawei Hong Kong Research Centre from June 2022 to May 2026. He was also the Director of Huawei Montreal Research Centre from March 2017 to Dec. 2020. Before that he was a principal researcher at Huawei Noah's Ark Lab Hong Kong from 2013 to 2017.

Dr. Geng holds a B.Eng. and M.Eng. in Electronic Engineering and Information Science from the University of Science and Technology of China (USTC), which he obtained in 2002 and 2005, respectively. He earned his Ph.D. degree in Electrical and Electronic Engineering from the University of Hong Kong (HKU) in 2009 under the supervision of Prof. Victor O.K. Li. Before joining Huawei, Dr. Geng served as a Post-Doctoral Research Fellow at HKU from 2009 to 2012. He also held the position of Senior Scientific Engineer at the Hong Kong Applied Science and Technology Research Institute (ASTRI) from 2012 to 2013.

His research interests include artificial intelligence, machine learning, information theory, wireless communication, and data-center networking. He has filed over 20 patents globally and has published more than 30 articles in prestigious journals and conferences, such as NeurIPS, AAAI, AISTATS, AAMAS, IEEE Transactions on Neural Networks and Learning Systems, ACM Sigcomm, IEEE/ACM Transactions on Networking, IEEE Transactions on Parallel and Distributed Systems, IEEE Infocom, ICNP, ICDCS, and others.

He has received numerous awards, including the IEEE ICC 2010 Best Paper Award, the Chinese Academy of Science President Award, the Excellent Postgraduate of Anhui Province Award, the Excellent Postgraduate of USTC Award, the USTC Outstanding Student Scholarship, the Lenovo Scholarship, and others. He is a senior member of IEEE and a member of ACM.

Yanhui Geng

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Education

- *The University of Hong Kong (Hong Kong)* – PhD in Electrical and Electronic Engineering, 2005-2009
Thesis "A framework to model the relationship between information and network performance", supervised by Prof. Victor O.K. Li (*Chair Professor of Information Engineering, IEEE Fellow*). Fully financed by HKU scholarship.
- *University of Science and Technology of China (Hefei, China)* – Master of Engineering and Bachelor of Engineering in Electronic Engineering and Information Science, 2002-2005 and 1998-2002

Experience

DIRECTOR, AI LAB, HUAWEI HONG KONG RESEARCH CENTRE - 2026/5-PRESENT

- My primary responsibility is to lead and manage the AI Lab of HKRC to conduct frontier research and development work in LLM, MLLM, AI infra, and next-generation AI paradigm.

DIRECTOR, HUAWEI HONG KONG RESEARCH CENTRE - 2022/6-2026/5

- As the Director of Huawei Hong Kong Research Centre, my primary responsibility is to enhance Huawei's research and development capabilities and foster a thriving research ecosystem in Hong Kong. This involves collaborating closely with esteemed universities and local industries to promote innovation and technological advancement.

DIRECTOR, HUAWEI NOAH'S ARK LAB HONG KONG - 2021/1-2022/5

- Noah's Ark Lab is Huawei's lab dedicated to fundamental and applied AI (Artificial Intelligence) research. Our mission is to push the frontier of AI research and build cutting-edge AI technologies used in the products and solutions of Huawei and our partners. I also help Huawei connect and build collaboration with leading universities in Hong Kong.

HEAD, HUAWEI MONTRÉAL RESEARCH CENTRE - 2017/3-2020/12

- As the director of Huawei Montréal Research Centre, I oversee both research and operational activities. My primary objective is to enhance Huawei's artificial intelligence (AI) capabilities by leveraging the expertise of top Canadian AI resources. Additionally, I am committed to fostering an AI ecosystem in Montréal, collaborating with local universities and industries to promote the development and adoption of AI technologies.

PRINCIPAL RESEARCHER, HUAWEI NOAH'S ARK LAB HONG KONG - 2013/7-2017/2

- Principal Investigator of a company-level project that focuses on designing the next generation of Software-Defined Networks (SDNs) by applying machine learning and data analytics to real-time network traffic. This approach enables the adoption of more intelligent network control strategies.

SENIOR SCIENTIFIC ENGINEER, HONG KONG APPLIED SCIENCE AND TECHNOLOGY RESEARCH INSTITUTE - 2012/2-2013/5

- Core member of the project "High-performance cloud computing system." I am responsible for the design and implementation of an Infrastructure as a Service (IaaS) cloud computing management platform. This platform provides high-performance computing resources and on-demand cloud computing services to small and medium enterprises and non-governmental organizations (NGOs) in Hong Kong.

POST-DOC RESEARCH FELLOW, THE UNIVERSITY OF HONG KONG – 2009/9-2012/1

- In the project titled “Low-cost versatile tracking device and technology for logistic applications,” I led the software and algorithm team, which was responsible for the design and implementation of a positioning engine utilizing a WiFi fingerprint matching algorithm.

Awards

- Huawei Award of Distinction and Dedication (top 1%), 2021
- Huawei 2012 Lab President’s Award (top 2%), 2020 and 2017
- Huawei Central Research Institute President’s Award (5%), 2018
- Huawei Excellent Supervisor Award (10%), 2017
- Huawei Future Star Award (top 20%), 2015
- Huawei Excellent Team Leader Award (top 5%), 2014
- IEEE ICC 2010 Best Paper Award (1 out of 600), 2010
- HKU Postgraduate Scholarship, 2005-2009
- Chinese Academy of Science President Award (1 out of 130), 2005
- Excellent Postgraduate Award of Anhui Province (2 out of 130), 2005
- Excellent Postgraduate Award of USTC (8 out of 130), 2005
- Lenovo Scholarship (1 out of 130), 2003
- USTC Outstanding Student Scholarship (30 out of 150), 2001 and 1998

Publications

1. Xiaofeng Liu, Qing Wang, Yunfeng Shao, Yanhui Geng and Yinchuan Li, “Structured Directional Pruning via Perturbation Orthogonal Projection,” IEEE Transactions on Signal Processing, November 2024.
2. Xiaofeng Liu, Yinchuan Li, Qing Wang, Xu Zhang, Yunfeng Shao and Yanhui Geng, “Sparse Personalized Federated Learning,” IEEE Transactions on Neural Networks and Learning Systems, February 2023.
3. Mehrtash Mehrabi, Walid Masoudimansour, Yingxue Zhang, Jie Chuai, Zhitang Chen, Mark Coates, Jianye Hao and Yanhui Geng, “Neighbor Auto-Grouping Graph Neural Networks for Handover Parameter Configuration in Cellular Network,” In Proceedings of Thirty-seven AAAI Conference on Artificial Intelligence (AAAI 2023), Washington DC, USA, February 2023.
4. Kaiyang Guo, Yunfeng shao and Yanhui Geng, “Model-Based Offline Reinforcement Learning with Pessimism-Modulated Dynamics Belief,” In Proceedings of Thirty-six Conference on Neural Information Processing Systems (NeurIPS 2022 Oral), New Orleans, USA, December 2022.
5. Junlong Lyu, Zhitang Chen, Chang Feng, Wenjing Cun, Shengyu Zhu, Yanhui Geng, Zhijie Xu and Yongwei Chen, “Para-CFlows: $\mathcal{C}^k$ -universal diffeomorphism approximators as superior neural surrogates,” In Proceedings of Thirty-six Conference on Neural Information Processing Systems (NeurIPS 2022), New Orleans, USA, December 2022.
6. Florence Regola, Soumyasundar Pal, Jianing Sun, Yingxue Zhang, Yanhui Geng and Mark Coates, “Node copying: A random graph model for effective graph sampling,” Elsevier Signal Processing Journal, vol. 192, March 2022.

7. Jiaqi Zheng, Hong Xu, Xiaojun Zhu, Guihai Chen and Yanhui Geng, "Sentinel: Failure Recovery in Centralized Traffic Engineering," IEEE/ACM Transactions on Networking, August 2019.
8. George Trimponias, Yan Xiao, Xiaorui Wu, Hong Xu and Yanhui Geng, "Node-Constrained Traffic Engineering: Theory and Applications," with segment routing in SDN based WANs," IEEE/ACM Transactions on Networking, June 2019.
9. Shoubo Hu, Bogdan Cautis, Zhitang Chen, Laiwan Chan, Yanhui Geng and Xiuqiang He, "Model-free inference of diffusion networks using RKHS embeddings," Springer Journal of Data Mining and Knowledge Discovery, March 2019.
10. Peng Wang, George Trimponias, Henry Xu and Yanhui Geng, "Luopan: sampling based load balancing in data center networks," IEEE Transactions on Parallel and Distributed Systems, January 2019.
11. Shoubo Hu, Zhitang Chen, Vahid Partovi Nia, Laiwan Chan and Yanhui Geng, "Causal Inference and Mechanism Clustering of a Mixture of Additive Noise Models," In Proceedings of Thirty-second Conference on Neural Information Processing Systems (NeurIPS 2018), Montreal, Canada, December 2018.
12. Xueying Gao, George Trimponias, Xiaoxiao Wang, Zhitang Chen, Yanhui Geng and Xin Liu, "Learning-based joint configuration for cellular networks," IEEE Internet of Things Journal, December 2018.
13. Zhuoshu Li, Zhitang Chen, Pascal Poupart, Sanmay Das and Yanhui Geng, "Faster policy adaptation in environment with exogeneity: a state augmentation approach," In Proceedings of International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2018), Stockholm, Sweden, July 2018.
14. Zhitang Chen, Ke He, Jian Li, Yanhui Geng, "Seq2Img: A sequence-to-image based approach towards IP traffic classification using convolutional neural networks," In Proceedings of IEEE International Conference on Big Data (Big Data 2017), Boston, USA, December 2017.
15. Xueying Gao, George Trimponias, Xiaoxiao Wang, Zhitang Chen, Yanhui Geng and Xin Liu, "Cellular network configuration via online learning and joint optimization," In Proceedings of IEEE International Conference on Big Data (Big Data 2017), Boston, USA, December 2017.
16. Jiaqi Zheng, Hong Xu, Xiaojun Zhu, Guihai Chen and Yanhui Geng, "We've got you covered: failure recovery with backup tunnels in traffic engineering," In Proceedings of IEEE International Conference on Network Protocols (IEEE ICNP 2016), Singapore, November 2016.
17. Zhitang Chen, Jiayao Wen and Yanhui Geng, "Predicting future traffic using hidden markov models," NetworkML 2016: the IEEE ICNP Workshop on Machine Learning in Computer Networks, Singapore, November 2016.
18. Pascal Poupart, Zhitang Chen, Priyank Jaini, Fred Fung, Hengky Susanto, Yanhui Geng, Li Chen, Kai Chen and Hao Jin, "Online flow size prediction for improved network routing," NetworkML 2016: the IEEE ICNP Workshop on Machine Learning in Computer Networks, Singapore, November 2016.
19. Peng Wang, George Trimponias, Henry Xu, Hongyuan Liu and Yanhui Geng, "Luopan: sampling based congestion-aware load balancing in data center networks," Poster at IEEE International Conference on Network Protocols (IEEE ICNP 2016), Singapore, November 2016.
20. Hong Zhang, Li Chen, Bairen Yi, Kai Chen, Mosharaf Chowdhury and Yanhui Geng, "CODA: towards automatically identifying and scheduling coflows in the dark," In Proceedings of ACM Sigcomm 2016, Florianópolis, Brazil, August 2016.
21. Zhitang Chen, Pascal Poupart and Yanhui Geng, "Online relative entropy search using reproducing kernel Hilbert space embeddings," In Proceedings of International Conference on Artificial Intelligence and Statistics (AISTATS 2016), Cadiz, Spain, May 2016.

22. Yang Peng, Kai Chen, Guohui Wang, Wei Bai, Yangming Zhao, Hao Wang, Yanhui Geng, Zhiqiang Ma and Lin Gu, "Towards comprehensive traffic forecasting in cloud computing: design and application," IEEE/ACM Transactions on Networking, August 2016.
23. Hao Wang, Li Chen, Kai Chen, Ziyang Li, Yiming Zhang, Haibing Guan, Zhengwei Qi, Dongsheng Li and Yanhui Geng, "FlowProphet: generic and accurate traffic prediction for data-parallel cluster computing," In Proceedings of IEEE International Conference on Distributed Computing Systems (IEEE ICDCS 2015), Columbus, USA, June 2015.
24. Yangming Zhao, Kai Chen, Wei Bai, Minlan Yu, Chen Tian, Yanhui Geng, Yiming Zhang, Dan Li and Sheng Wang, "RAPIER: integrating routing and scheduling for coflow-aware data center networks," In Proceedings of IEEE Conference on Computer Communications (IEEE INFOCOM 2015), Hong Kong, April 2015.
25. Shrikanth Narayanan, Ayush Jaiswal, Yao-Yi Chiang, Yanhui Geng, Craig A. Knoblock and Pedro Szekely, "Integration and automation of data preparation and data mining," In Proceedings of IEEE International Conference on Data Mining 2014 (IEEE ICDM 2014), Shenzhen, China, December 2014.
26. Albert Y.S. Lam, Yanhui Geng, and Victor O.K. Li, "Information-theoretic bounds for performance of resource-constrained communication systems," arXiv (1404.0142), April 2014.
27. Yanhui Geng, Albert Y.S. Lam and Victor O.K. Li, "Performance bounds of opportunistic scheduling in wireless networks," In Proceedings of IEEE Global Communications Conference (IEEE GLOBECOM 2010), Miami, USA, December 2010.
28. Yanhui Geng, Albert Y.S. Lam and Victor O.K. Li, "An information-theoretic model for resource-constrained systems," In Proceedings of IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2010), Istanbul, Turkey, October 2010.
29. Yanhui Geng, Victor O.K. Li, "A framework to analyze network performance based on information quality," In Proceedings of IEEE International Conference on Communications (IEEE ICC 2010), Cape Town, South Africa, May 2010.
30. Yanhui Geng, Victor O.K. Li, "Mathematical impact of information accuracy on network performance," In Proceedings of IEEE International Conference on Communications (IEEE ICC 2010), Cape Town, South Africa, May 2010.
31. Yanhui Geng, Qiong Sun, Yuzhong Chen and Nenghai Yu, "A novel approach to improve the performance of MPLS-based VPN," In Proceedings of IEEE International Conference on Intelligent Engineering Systems (IEEE INES 2004), Cluj Napoca, Romania, September 2004.
32. Yanhui Geng, Qiong Sun and Nenghai Yu, "A mechanism applied to improving the performance of MPLS VPN," In Proceedings of IEEE 8th Russian-Korean International Symposium on Science and Technologies (IEEE KORUS 2004), Tomsk, Russia, June 2004.
33. Yanhui Geng, Nenghai Yu, Bingqing Yang and Yuzhong Chen, "An analysis and comparison of L2 link failure solutions in MPLS networks," In Proceedings of IEEE International Conference on Intelligent Engineering Systems (IEEE INES 2003), Assiut-Luxor, Egypt, March 2003.
34. Qiong Sun, Yanhui Geng, Hui Wang and Nenghai Yu, "A multipath mechanism applied to traffic engineering in MPLS network," In Proceedings of IEEE International Conference on Intelligent Engineering Systems (IEEE INES 2004), Cluj Napoca, Romania, September 2004.
35. Qiong Sun, Yanhui Geng, , Hui Wang and Nenghai Yu, "Reordering on internet based on metric," In Proceedings of IEEE 8th Russian-Korean International Symposium on Science and Technologies (IEEE KORUS 2004), Tomsk, Russia, June 2004.
36. Qiong Sun, Yanhui Geng and Nenghai Yu, "A mechanism to improve the performance of MPLS-VPN based on using label stack," Journal of Computer Science, no. 2, 2006.

37. Yuzhong Chen, Caihong Kai, Yanhui Geng and Feng Li, "Performance analysis and improvement of IEEE 802.11 WLAN," In Proceedings of IEEE Asia-Pacific Conference on Communications (IEEE APCC 2004), Beijing, China, August 2004.
38. Yuzhong Chen, Nenghai Yu, Yanhui Geng and Xuelong Li, "A traffic engineering scheme for Diffserv over MPLS," In Proceedings of IEEE International Conference on Neural Networks and Signal Processing, (IEEE ICNNSP 2003), Nanjing, China, December 2003.
39. Yuzhong Chen, Nenghai Yu, Yanhui Geng and Xuelong Li, "Combination of traffic engineering and Diffserv over MPLS," In Proceedings of IEEE International Conference on Intelligent Engineering Systems (IEEE INES 2003), Assiut-Luxor, Egypt, March 2003.
40. Yuzhong Chen, Bingqing Yang, Rong Ren, Yanhui Geng and Nenghai Yu, "Traffic engineering with constraint-based routing in DiffServ/MPLS network," In Proceedings of IEEE Workshop on Local and Metropolitan Area Networks (IEEE LANMAN 2004), Mill Valley, USA, April 2004.

Patents

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1. Qibin Wu, Yanhui Geng and Zhitang Chen, "A method and apparatus for accelerated survival analysis based on spatial-temporal data mining from optical fiber link failure records", China Patent Application, #201710035513.X, September 2017.
 2. Zhitang Chen, Yanhui Geng and Georgios Trimponias, "A method and apparatus for network flow application type classification", China Patent Application, to be filed, #201610842056.0, June 2016.
 3. Georgios Trimponias, Zhitang Chen and Yanhui Geng, "Routing control in Software-Defined Networking by deep reinforcement learning", China Patent Application, filed, #201610465236.1, June 2016.
 4. Bing Ni and Yanhui Geng, "A visualization apparatus to describe bi-directional linked heat-map", Patent Cooperation Treaty, filed, #PCT/CN2015/091852, October 2015.
 5. Zhitang Chen, Yanhui Geng and Hong Zhang, "A method and apparatus for co-flow identification", China Patent Application, filed, #201610141226.2, March 2016.
 6. Yanhui Geng, Zhitang Chen, and Baofeng Zhang, "A self-learning and adaptive network control system and apparatus", China Patent Application, filed, #201510524510.3, August 2015.
 7. Zhitang Chen, Fred Fung and Yanhui Geng, "A distributed flow classification system and apparatus based on the switch flow pipeline", China Patent Application, filed, #201510522003.6, August 2015.
 8. Zhitang Chen, Baofeng Zhang and Yanhui Geng, "Adaptive fault-tolerant network control system", China Patent Application, filed, #201510447521.6, July 2015.
 9. Zhitang Chen, Yanhui Geng and Pascal Poupart, "Network flow prediction based on pattern similarities", China Patent Application, filed, #201510003825.3, January 2015.
 10. Yanhui Geng, Ko Yu Ting and Zhuolin Jiang, "A fast indoor positioning solution for smartphones based on image recognition", China Patent Application, filed, #201410549080.6, October 2014.
 11. Yanhui Geng, Kai Chen, Qiang Yang, "A traffic-engineering solution in SDN based on data mining", China Patent Application, filed, #201410253610.2, June 2014.

Professional Services

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- Keynote speaker at IEEE ICC 2022 Industry Presentations "Evolution of edge computing and edge AI towards 6G. Seoul, South Korea, May 2022.
 - Invited speaker at "Big Data Technology Conference". Beijing, China, December 2016.

- Program Co-Chair of the IEEE ICNP Workshop on Machine Learning in Computer Networks (NetworkML 2016). Singapore, November 2016.
- Invited talk titled "MIND: A machine learning based SDN control paradigm" at Open Networking Summit 2016. Santa Clara, US, March 2016.
- Keynote speaker at "Big-Data in Network Workshop", held by Tsinghua-UC Berkeley Shenzhen Institute (TBSI). Shenzhen, China, July 2015.
- Keynote speaker at "Multimedia, Big-data and Network Workshop", held by Tsinghua University. Shenzhen, China, September 2014.

Language

- Fluent English and Chinese, moderate Cantonese.